Current	Result	Size	Time	Cycles	GPU	SM Frequency	Process	Attributes
Current	922 - sm80_xmma_fprop_implicit_gemm_f32f32_f32f32_f32_nchwkcrs_nchw_tilesize32x32x8_stage3_warpsize1x2x1_g1_ffma_t1r3s3_aligna4_alignc4_execute_kernel_trt	(1, 65280, 1)x(64, 1, 1)	1.93 ms	4,802,547	0 - NVIDIA GeForce RTX 5060	2.49 Ghz	[27420] trtexec.exe	GPU Compute Capability: 12.0 Registers/thread: 90 # SMs: 30 # TPCs: 15

Summary | Details | Source | Context | Comments | Raw | Session

Compare | Tools | View | Export

GPU Speed Of Light Throughput

GPU Throughput Breakdown

High-level overview of the throughput for compute and memory resources of the GPU. For each unit, the throughput reports the achieved percentage of utilization with respect to the theoretical maximum. Breakdowns show the throughput for each individual sub-metric of Compute and Memory to clearly identify the highest contributor. High-level overview of the utilization for compute and memory resources of the GPU presented as a roofline chart.

Compute (SM) Throughput [%]	86.95	Duration [ms]	1.93
Memory Throughput [%]	78.29	Elapsed Cycles [cycle]	4,802,547
L1/TEX Cache Throughput [%]	78.40	SM Active Cycles [cycle]	4,792,103.87
L2 Cache Throughput [%]	38.83	SM Frequency [Ghz]	2.49
DRAM Throughput [%]	39.15	DRAM Frequency [Ghz]	13.79

High Throughput This workload is utilizing greater than 80.0% of the available compute or memory performance of this device. To further improve performance, work will likely need to be shifted from the most utilized to another unit. Start by analyzing workloads in the Compute Workload Analysis section.

Roofline Analysis The ratio of peak float (FP32) to double (FP64) performance on this device is 64:1. The workload achieved 32% of this device's FP32 peak performance and 0% of its FP64 peak performance. See the Profiling Guide for more details on roofline analysis.

Compute Throughput Breakdown

SM: Inst Executed [%]	86.95
SM: Issue Active [%]	86.95
SM: Inst Executed Pipe Lsu [%]	63.49
SM: Pipe Fmaheavy Cycles Active [%]	62.25
SM: Pipe Fma Cycles Active [%]	38.73
SM: Pipe Aluheavy Cycles Active [%]	37.52
SM: Mio Inst Issued [%]	29.81
SM: Inst Executed Pipe Adu [%]	25.94
IDC: Request Cycles Active [%]	21.59
SM: Mio2rf Writeback Active [%]	15.55
SM: Mio Pq Write Cycles Active [%]	7.98
SM: Mio Pq Read Cycles Active [%]	7.98
SM: Inst Executed Pipe Uniform [%]	5.87
SM: Memory Throughput Internal Activity [%]	4.72
SM: Inst Executed Pipe Cbu Pred On Any [%]	0.27
SM: Inst Executed Pipe Ipa [%]	0
SM: Inst Executed Pipe Tex [%]	0
SM: Inst Executed Pipe Xu [%]	0
SM: Instruction Throughput Internal Activity [%]	0
SM: Pipe Fp64 Cycles Active [%]	0
SM: Pipe Tensor Cycles Active [%]	0
SM: Pipe Tma Cycles Active [%]	0

Memory Throughput Breakdown

L1: Data Pipe Lsu Wavefronts [%]	78.29
L1: Lsuin Requests [%]	63.49
L1: Lsu Writeback Active [%]	51.42
DRAM: Cycles Active [%]	39.15
L2: Xbar2lts Cycles Active [%]	38.83
DRAM: Dram Sectors [%]	30.24
L2: T Tag Requests [%]	29.26
GPU: Compute Memory Access Throughput Internal Activity [%]	28.86
L2: T Sectors [%]	28.72
L2: Lts2xbar Cycles Active [%]	19.67
L2: D Sectors [%]	19.03
L1: Data Bank Reads [%]	17.34
L1: M Xbar2l1tex Read Sectors [%]	14.08
L1: M L1tex2xbar Req Cycles Active [%]	12.24
L1: Data Bank Writes [%]	7.92
L2: D Sectors Fill Device [%]	4.55
GPU: Compute Memory Request Throughput Internal Activity [%]	0.01
SYSLTS: Xbar2syslts Cycles Active [%]	0.01
SYSLTS: T Sectors [%]	0.01
SYSLTS: T Tag Requests [%]	0.01
SYSLTS: Syslts2xbar Cycles Active [%]	0.01
SYSLTS: D Sectors Fill Sysmem [%]	0.01
SYSLTS: D Sectors [%]	0.00
L1: Texin Sm2tex Req Cycles Active [%]	0.00
L1: Data Pipe Tex Wavefronts [%]	0
L1: F Wavefronts [%]	0
L1: M Xbar2l1tex Read Sectors Mem Dshared [%]	0
L1: Tex Writeback Active [%]	0
L1: Tmain Requests [%]	0
L2: D Atomic Input Cycles Active [%]	0
L2: D Decomp Input Sectors [%]	0
SYSLTS: D Atomic Input Cycles Active [%]	0

Compute Workload Analysis

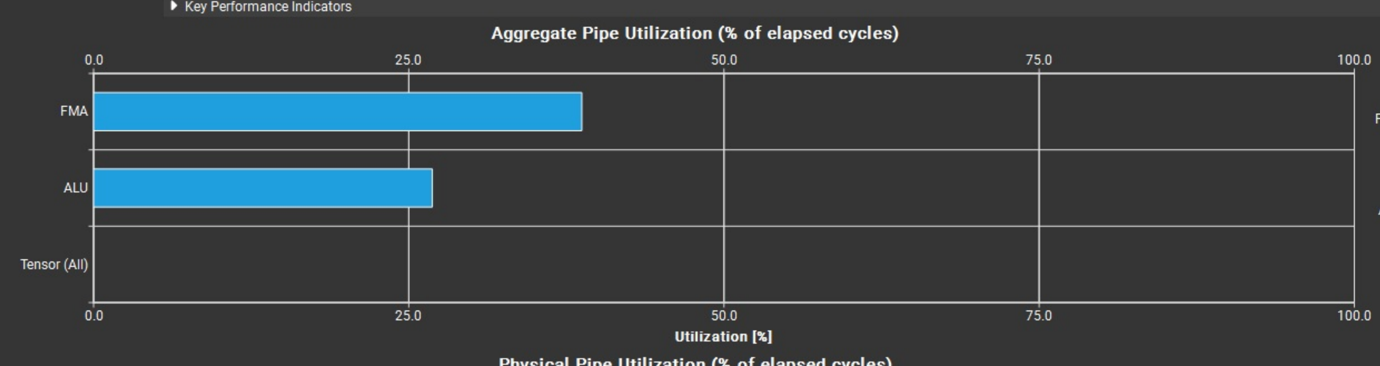
Pipe Utilization (Elapsed Cycles)

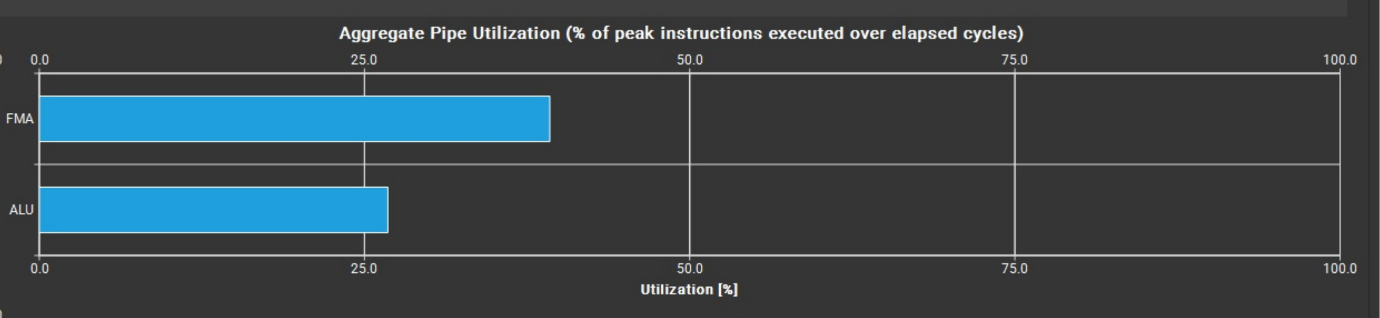
Detailed analysis of the compute resources of the streaming multiprocessors (SM), including the achieved instructions per clock (IPC) and the utilization of each available pipeline. Pipelines with very high utilization might limit the overall performance.

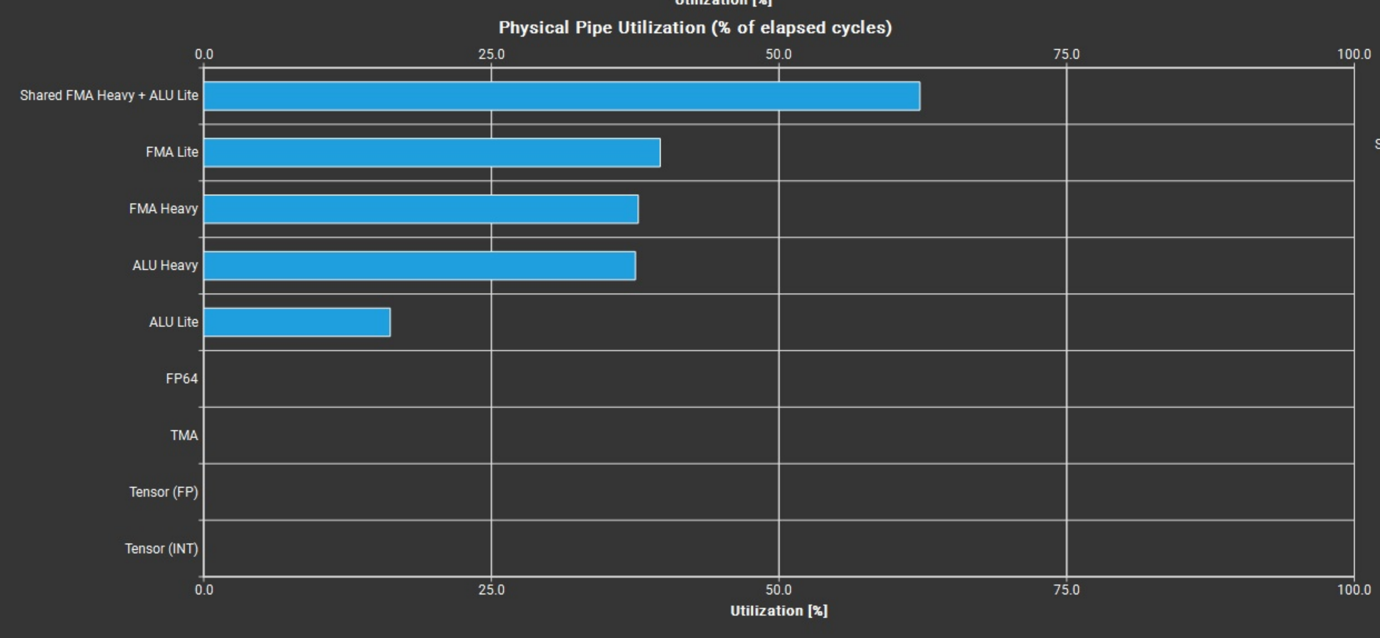
Executed Ipc Elapsed [inst/cycle]	3.48	SM Busy [%]	86.95
Executed Ipc Active [inst/cycle]	3.48	Issue Slots Busy [%]	86.95
Issued Ipc Active [inst/cycle]	3.48		

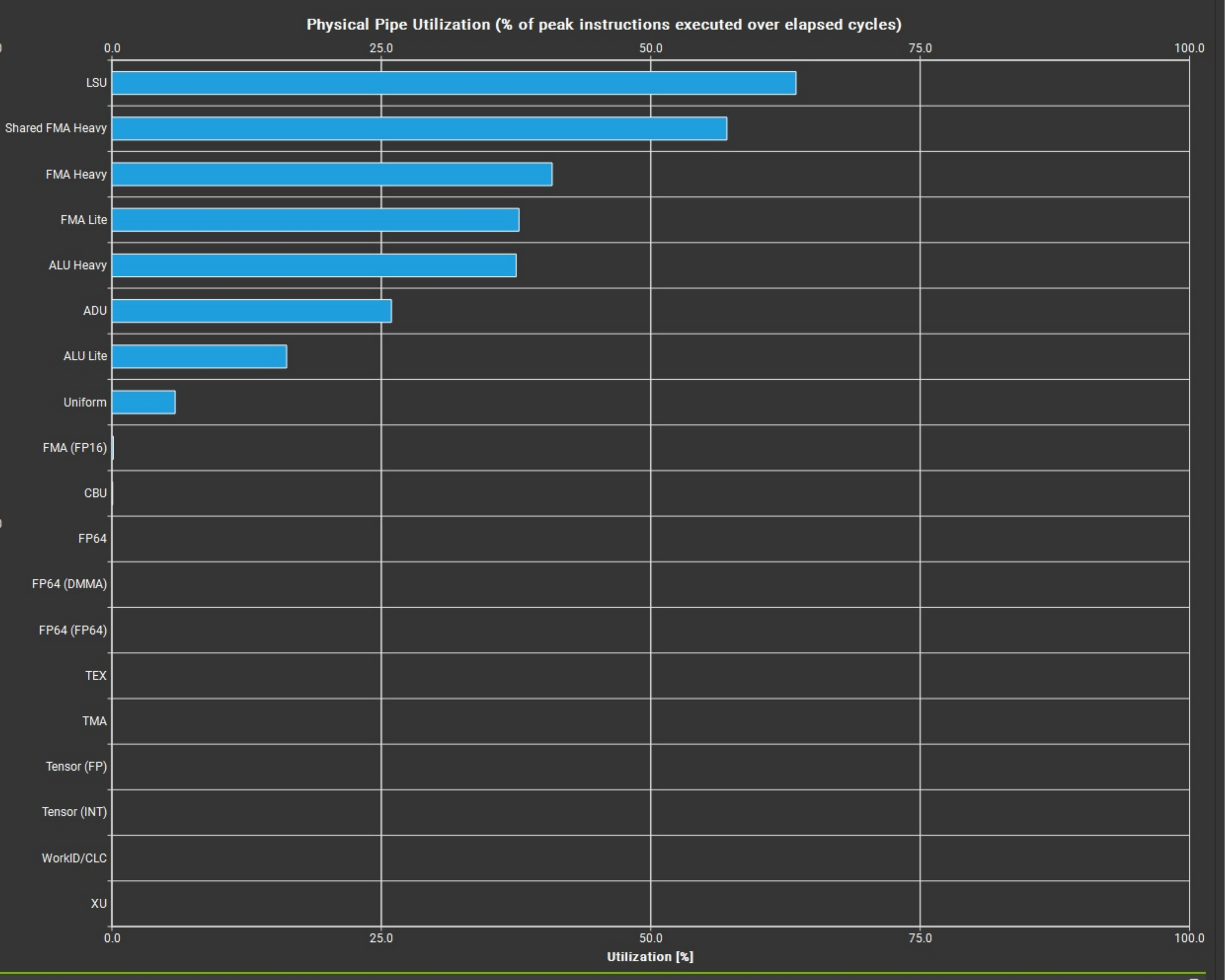
High Utilization Shared FMA Heavy is the highest-utilized pipeline (62.2%) based on elapsed cycles in the workload, taking into account the rates of its different instructions. It is a physical pipe and shared by the logical pipes FMA Heavy and ALU Lite. It's dominated by its FMA Heavy sub-pipeline. The pipeline is well-utilized, but might become a bottleneck if more work is added. Based on the number of executed instructions, the highest utilized pipeline (63.5%) is LSU. It executes load/store memory operations. Comparing the two, the overall pipeline utilization appears to be caused by frequent, low-latency instructions. See the Profiling Guide or hover over the pipeline name to understand the workloads handled by each pipeline. The Instruction Statistics section shows the mix of executed instructions for this workload.

Key Performance Indicators









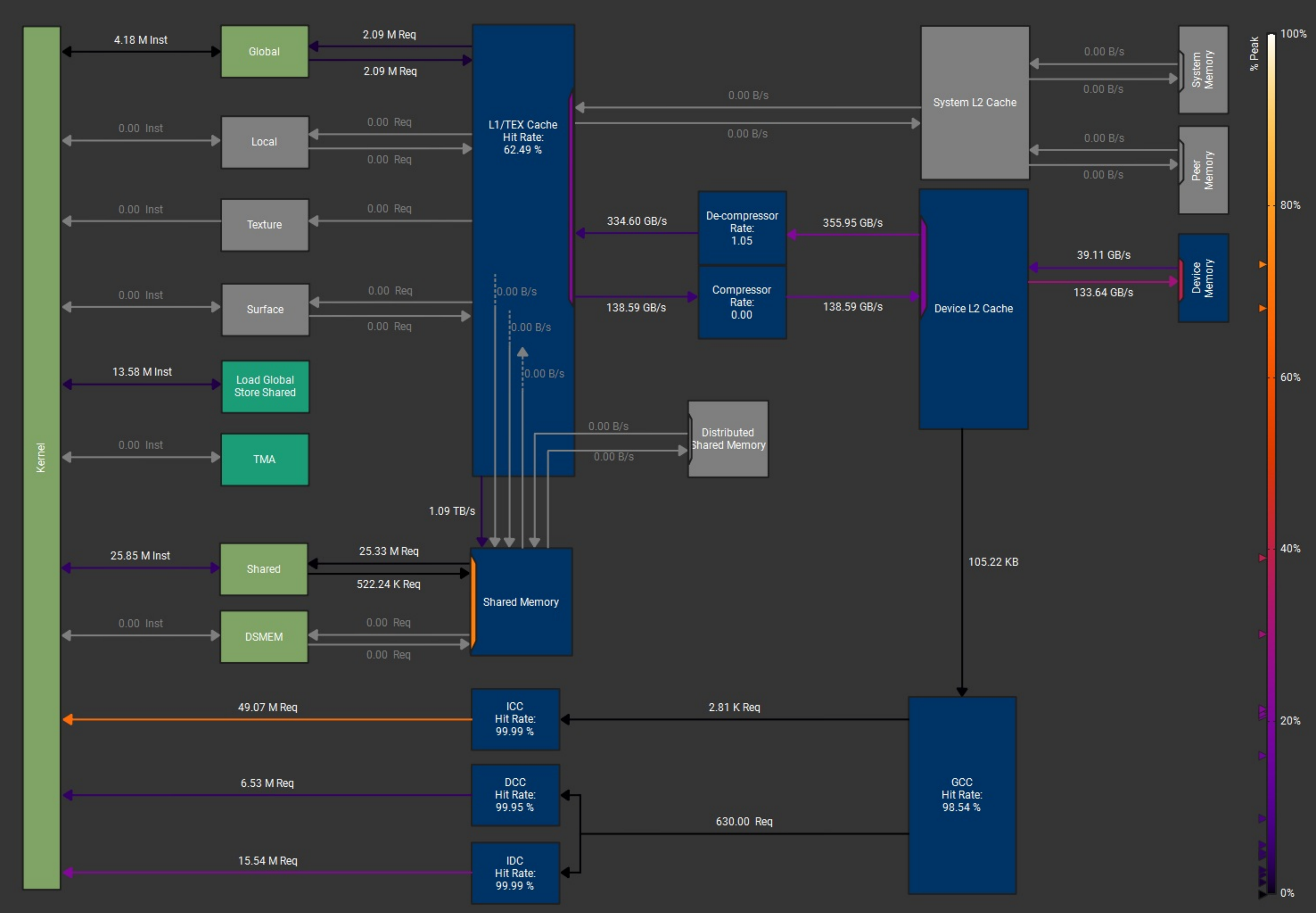
Memory Workload Analysis

Detailed analysis of the memory resources of the GPU. Memory can become a limiting factor for the overall kernel performance when fully utilizing the involved hardware units (Mem Busy), exhausting the available communication bandwidth between those units (Max Bandwidth), or by reaching the maximum throughput of issuing memory instructions (Mem Pipes Busy). Detailed chart of the memory units.

Memory Throughput [Gbyte/s]	172.75	Mem Busy [%]	78.29
L1/TEX Hit Rate [%]	62.49	Max Bandwidth [%]	63.49
L2 Hit Rate [%]	64.13	Mem Pipes Busy [%]	63.49
L2 Compression Input Sectors [sector]	0	Local Memory Spilling Requests	0
L2 Compression Ratio	0	Shared Memory Spilling Requests	0
L2 Compression Success Rate [%]	0	Local Memory Spilling Request Overhead [%]	0
L2 Persisting Size [Mbyte]	4.72	Shared Memory Spilling Request Overhead [%]	0
L2 Sector Promotion Misses [%]	9.31	-	-

Memory Chart

Values: Throughput Inactivity: Greyed Out



Launch Statistics

Summary of the configuration used to launch the kernel. The launch configuration defines the size of the kernel grid, the division of the grid into blocks, and the GPU resources needed to execute the kernel. Choosing an efficient launch configuration maximizes device utilization.

Grid Size	65,280	Function Cache Configuration	CachePreferNone
Cluster Size	0	Preferred Cluster Size	0
Registers Per Thread [register/thread]	90	Cluster Scheduling Policy	PolicySpread
Static Shared Memory Per Block [byte/block]	0	Block Size	64
Dynamic Shared Memory Per Block [Kbyte/block]	6.53	Threads [thread]	4,177,920
Driver Shared Memory Per Block [Kbyte/block]	1.02	Waves Per SM	217.60
Shared Memory Configuration Size [Kbyte]	102.40	Uses Green Context	0
Stack Size	1,024	# SMs [SM]	30
# TPCs	15	Enabled TPC IDs	all

WorkID/CLC Info

	Average	Min	Max	Sum
WorkId/CLC Requests	0	0	0	0
WorkId/CLC Requests Granted	0	0	0	0
WorkId/CLC Requests Granted as CTAs	0	0	0	0

Occupancy

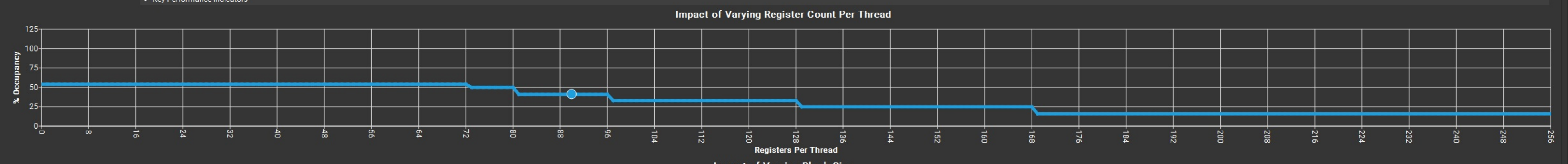
% Occupancy Graphs

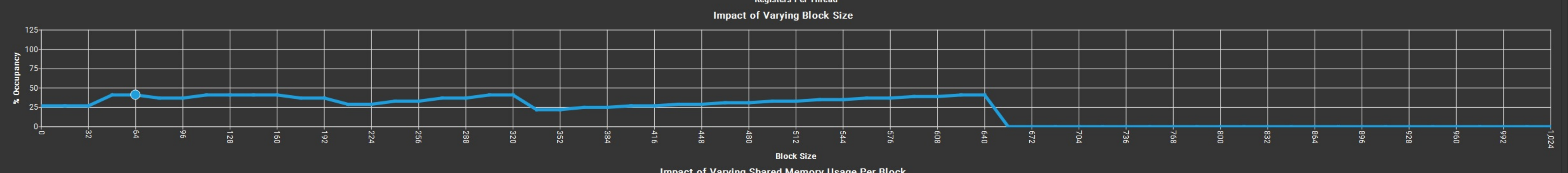
Occupancy is the ratio of the number of active warps per multiprocessor to the maximum number of possible active warps. Another way to view occupancy is the percentage of the hardware's ability to process warps that is actively in use. Higher occupancy does not always result in higher performance, however, low occupancy always reduces the ability to hide latencies, resulting in overall performance degradation. Large discrepancies between the theoretical and the achieved occupancy during execution typically indicates highly imbalanced workloads.

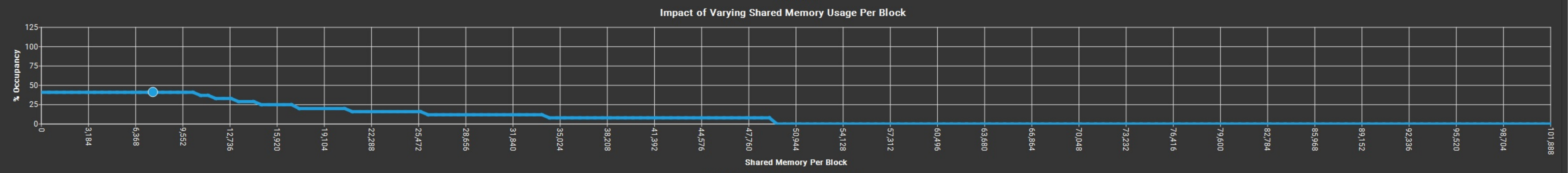
Theoretical Occupancy [%]	41.67	Block Limit Registers [block]	10
Theoretical Active Warps per SM [warp]	20	Block Limit Shared Mem [block]	13
Achieved Occupancy [%]	40.96	Block Limit Warps [block]	24
Achieved Active Warps Per SM [warp]	19.66	Block Limit SM [block]	24
Cluster Occupancy [%]	0	Block Limit Barriers [block]	24
Max Active Clusters [cluster]	0	Max Cluster Size [block]	8
Overall GPU Occupancy [%]	0		

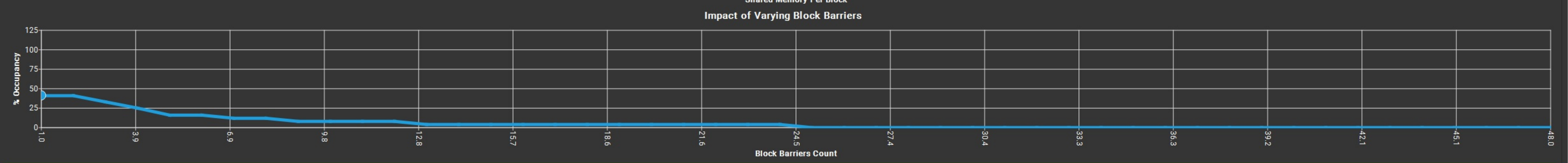
Theoretical Occupancy
Est. Local Speedup: 58.33% The 5.00 theoretical warps per scheduler this kernel can issue according to its occupancy are below the hardware maximum of 12. This kernel's theoretical occupancy (41.7%) is limited by the number of required registers.

Key Performance Indicators









GPU and Memory Workload Distribution

Analysis of workload distribution in active cycles of SM, SMP, SMSP, L1 & L2 caches, and DRAM

Average SM Active Cycles [cycle]	4,792,103.87	Average L1 Active Cycles [cycle]	4,792,103.87
Average L2 Active Cycles [cycle]	4,302,468.50	Average SMSP Active Cycles [cycle]	4,792,299.92
Average DRAM Active Cycles [cycle]	10,415,560	Total SM Elapsed Cycles [cycle]	143,955,440
Total L1 Elapsed Cycles [cycle]	143,955,440	Total L2 Elapsed Cycles [cycle]	51,663,552
Total SMSP Elapsed Cycles [cycle]	575,821,760	Total DRAM Elapsed Cycles [cycle]	106,409,984

Workload Distribution

	Average	Min	Max	Sum
SM Active Cycles	4,792,103.87	4,786,696	4,796,491	143,763,116
SMSP Active Cycles	4,792,299.92	4,785,891	4,796,612	575,075,991
L1 Active Cycles	4,792,103.87	4,786,696	4,796,491	143,763,116
L2 Active Cycles	4,302,468.50	4,301,858	4,302,905	51,629,622
DRAM Active Cycles	10,415,560	10,415,200	10,415,968	41,662,240

Source Counters

Source metrics, including branch efficiency and sampled warp stall reasons. Warp Stall Sampling metrics are periodically sampled over the kernel runtime. They indicate when warps were stalled and couldn't be scheduled. See the documentation for a description of all stall reasons. Only focus on stalls if the schedulers fail to issue every cycle.

Branch Instructions [inst]	4,177,920	Branch Efficiency [%]	100
Branch Instructions Ratio [%]	0.01	Avg. Divergent Branches [branches]	0

Uncoalesced Global Accesses
Est. Speedup: 30.53% This workload has uncoalesced global accesses resulting in a total of 23309364 excessive sectors (31% of the total 76303764 sectors). Check the L2 Theoretical Sectors Global Excessive table for the primary source locations. The CUDA Programming Guide has additional information on reducing uncoalesced device memory accesses.

Key Performance Indicators

Metric Name	Value	Guidance
derived__memory_l2_theoretical_sectors_global_excessive	2.33094E+07	Reduce the number of excessive wavefronts in L2

L2 Theoretical Sectors Global Excessive

Location	Value	Value (%)
0xb011786c0 in sm80_xmma_fprop_implicit_gemm_f32f32_f32f32_f32_nchwkcrs_nchw_tilesize32x32x8_stage3_warpsize1x2x1_g1_ffma_t1r3s...	4,047,360	17
0xb01178530 in sm80_xmma_fprop_implicit_gemm_f32f32_f32f32_f32_nchwkcrs_nchw_tilesize32x32x8_stage3_warpsize1x2x1_g1_ffma_t1r3s...	4,047,360	17
0xb01178490 in sm80_xmma_fprop_implicit_gemm_f32f32_f32f32_f32_nchwkcrs_nchw_tilesize32x32x8_stage3_warpsize1x2x1_g1_ffma_t1r3s...	4,047,360	17
0xb01178450 in sm80_xmma_fprop_implicit_gemm_f32f32_f32f32_f32_nchwkcrs_nchw_tilesize32x32x8_stage3_warpsize1x2x1_g1_ffma_t1r3s...	4,047,360	17
0xb01177750 in sm80_xmma_fprop_implicit_gemm_f32f32_f32f32_f32_nchwkcrs_nchw_tilesize32x32x8_stage3_warpsize1x2x1_g1_ffma_t1r3s...	913,920	4

Uncoalesced Shared Accesses
Est. Speedup: 7.70% This workload has uncoalesced shared accesses resulting in a total of 7311360 excessive wavefronts (8% of the total 94786560 wavefronts). Check the L1 Wavefronts Shared Excessive table for the primary source locations. The CUDA Best Practices Guide has an example on optimizing shared memory accesses.

Key Performance Indicators

Metric Name	Value	Guidance
derived__memory_l1_wavefronts_shared_excessive	7.31136E+06	Reduce the number of excessive wavefronts in L1TEX

L1 Wavefronts Shared Excessive

Location	Value	Value (%)
0xb011786c0 in sm80_xmma_fprop_implicit_gemm_f32f32_f32f32_f32_nchwkcrs_nchw_tilesize32x32x8_stage3_warpsize1x2x1_g1_ffma_t1r3s...	1,044,480	14
0xb01178530 in sm80_xmma_fprop_implicit_gemm_f32f32_f32f32_f32_nchwkcrs_nchw_tilesize32x32x8_stage3_warpsize1x2x1_g1_ffma_t1r3s...	1,044,480	14
0xb01178490 in sm80_xmma_fprop_implicit_gemm_f32f32_f32f32_f32_nchwkcrs_nchw_tilesize32x32x8_stage3_warpsize1x2x1_g1_ffma_t1r3s...	1,044,480	14
0xb01178450 in sm80_xmma_fprop_implicit_gemm_f32f32_f32f32_f32_nchwkcrs_nchw_tilesize32x32x8_stage3_warpsize1x2x1_g1_ffma_t1r3s...	1,044,480	14
0xb0117c6a0 in sm80_xmma_fprop_implicit_gemm_f32f32_f32f32_f32_nchwkcrs_nchw_tilesize32x32x8_stage3_warpsize1x2x1_g1_ffma_t1r3s...	522,240	7

Warp Stall Sampling (Not-issued Samples)

Location	Value	Value (%)
0xb01178ac0 in sm80_xmma_fprop_implicit_gemm_f32f32_f32f32_...	316	5
0xb011774a0 in sm80_xmma_fprop_implicit_gemm_f32f32_f32f32_...	290	5
0xb01178720 in sm80_xmma_fprop_implicit_gemm_f32f32_f32f32_...	211	3
0xb01178750 in sm80_xmma_fprop_implicit_gemm_f32f32_f32f32_...	203	3
0xb01177dc0 in sm80_xmma_fprop_implicit_gemm_f32f32_f32f32_...	196	3

Most Instructions Executed

Location	Value	Value (%)
0xb01178e20 in sm80_xmma_fprop_implicit_gemm_f32f32_f32f32_...	1,436,160	0
0xb01178e10 in sm80_xmma_fprop_implicit_gemm_f32f32_f32f32_...	1,436,160	0
0xb01178e00 in sm80_xmma_fprop_implicit_gemm_f32f32_f32f32_...	1,436,160	0
0xb01178df0 in sm80_xmma_fprop_implicit_gemm_f32f32_f32f32_f...	1,436,160	0
0xb01178de0 in sm80_xmma_fprop_implicit_gemm_f32f32_f32f32_...	1,436,160	0

Still missing data? Collect the *full* metric set, enable individual sections or learn how to select individual metrics for collection.